CPU Scheduling

Topic 4

CPU Scheduling

CPU Scheduling is the process by which an operating system selects one process from the ready queue and assigns the CPU to it for execution.

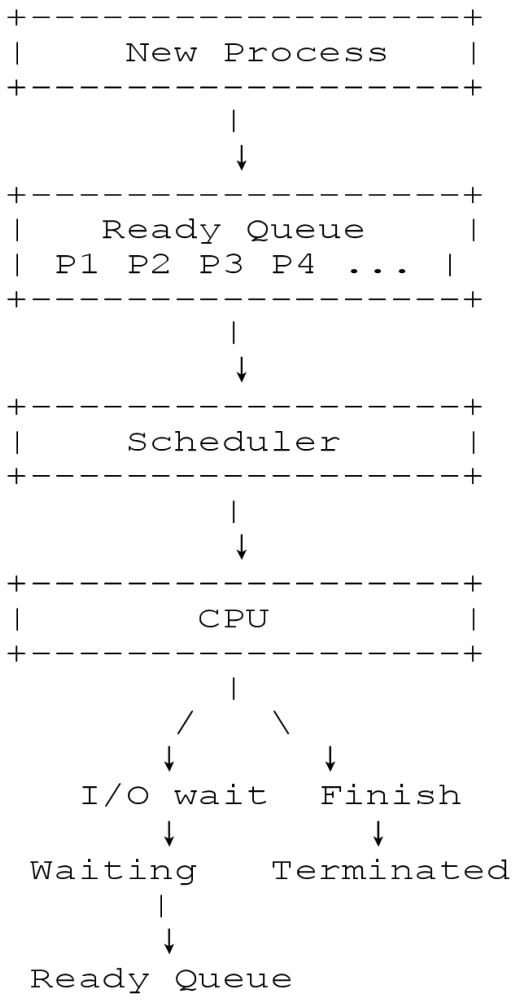
Since several processes may compete for the CPU at the same time, scheduling helps the operating system use the processor efficiently and provide good system performance.

1. Objectives of CPU Scheduling

The main objectives are to:

- Keep the CPU as busy as possible.
- Maximize the number of processes completed per unit of time.
- Minimize the time a process spends waiting.
- Minimize the total time taken to complete a process.
- Provide quick response to interactive users.
- Ensure fair allocation of CPU time.
- Maintain predictable and efficient system performance.

2. Basic CPU Scheduling Process



The **CPU scheduler** selects a process from the ready queue according to a scheduling algorithm.

3. Types of CPU Scheduling

A. Preemptive Scheduling

In preemptive scheduling, the operating system can interrupt a running process and assign the CPU to another process.

Examples:

- Round Robin
- Shortest Remaining Time First
- Preemptive Priority Scheduling

Advantage: Good response time.

Disadvantage: Context-switching overhead can be high.

Types of CPU Scheduling

B. Non-Preemptive Scheduling

In non-preemptive scheduling, once a process receives the CPU, it normally keeps it until it terminates or enters a waiting state.

Examples:

- FCFS
- Non-preemptive SJF
- Non-preemptive Priority Scheduling

Advantage: Simple and less context-switching overhead.

Disadvantage: A long process can make shorter processes wait.

First Come First Serve (FCFS)

- In FCFS Scheduling
- The process which arrives first in the ready queue is firstly assigned the CPU.
- In case of a tie, process with smaller process id is executed first.
- It is always non-preemptive in nature.
- Jobs are executed on first come, first serve basis.
- It is a non-preemptive, pre-emptive scheduling algorithm.
- Easy to understand and implement.
- Its implementation is based on FIFO queue.
- Poor in performance as average wait time is high.

First Come First Serve (FCFS)

Advantages-

- It is simple and easy to understand.
- It can be easily implemented using queue data structure.
- It does not lead to starvation.

Disadvantages-

- It does not consider the priority or burst time of the processes.
- It suffers from convoy effect i.e. processes with higher burst time arrived before
- the processes with smaller burst time.

4. CPU Scheduling Algorithms

4.1 First-Come, First-Served (FCFS)

FCFS schedules processes according to their arrival order.

The process that arrives first gets the CPU first.

Example

Suppose:

Process	Arrival Time	Burst Time
P1	0	5
P2	1	3
P3	2	2

The execution order is:

P1 → P2 → P3

First-Come, First-Served (FCFS)

Gantt Chart

0 5 8 10

|--- P1 ---|-- P2 --|--P3--|

- **Advantages**

Very simple.

Easy to implement.

Fair according to arrival order.

- **Disadvantages**

Can produce long waiting times.

Suffers from the convoy effect, where short processes wait behind a long process.

5. Shortest Job First (SJF)

Process which have the shortest burst time are scheduled first.

- If two processes have the same bust time, then FCFS is used to break the tie.
- This is a non-pre-emptive, pre-emptive scheduling algorithm.
- Best approach to minimize waiting time.
- Easy to implement in Batch systems where required CPU time is known in advance.
- Impossible to implement in interactive systems where required CPU time is not known.
- The processer should know in advance how much time process will take.
- Pre-emptive mode of Shortest Job First is called as Shortest Remaining Time First (SRTF).

5. Shortest Job First (SJF)

SJF selects the process with the shortest CPU burst time.

Example:

Process	Burst Time
P1	8
P2	3
P3	5
P4	2

Execution order:

P4 → P2 → P3 → P1

5. Shortest Job First (SJF)

- **Advantages**

  Minimizes average waiting time when burst times are known accurately.

  Good for batch processing.

- **Disadvantages**

  CPU burst time is difficult to know in advance.

  Long processes may experience starvation.

6. Shortest Remaining Time First (SRTF)

SRTF is the preemptive version of SJF.

The process with the shortest remaining execution time gets the CPU.

If a new process arrives with a shorter remaining time than the currently running process, the current process can be interrupted.

Example

P1 running

↓

P2 arrives with shorter remaining time

↓

P1 is preempted

↓

P2 runs

Shortest Remaining Time First (SRTF)

Advantages

- Provides good average waiting time.
- Gives preference to short processes.

Disadvantages

- More context switching.
- Can cause starvation of long processes.
- Requires estimates of remaining burst time.

7. Priority Scheduling

In **Priority Scheduling**, each process is assigned a priority. The CPU is allocated to the process with the highest priority.

For example:

Process	Priority
P1	3
P2	1
P3	2

If **1 represents the highest priority**, the execution order is:

`P2 → P3 → P1`

- **Priority scheduling can be:**

  Preemptive

  Non-preemptive

Problem: Starvation

A low-priority process may wait for a very long time because higher-priority processes continually arrive.

Solution: Aging

Aging gradually increases the priority of a waiting process so that it eventually receives CPU time.

8. Round Robin (RR)

Round Robin is a preemptive scheduling algorithm commonly used in time-sharing systems.

Each process receives a fixed amount of CPU time called a time quantum or time slice.

For example, if the time quantum is 3 milliseconds:

- P1 → P2 → P3 → P1 → P2 → P3 ...

Example

Suppose:

Process	Burst Time
P1	5
P2	4
P3	2

Time quantum = **2**

A possible execution sequence is:

P1 → P2 → P3 → P1 → P2 → P1

Round Robin (RR)

Advantages

- Fair to processes.
- Good response time.
- Suitable for interactive and time-sharing systems.
- Prevents a single process from monopolizing the CPU.

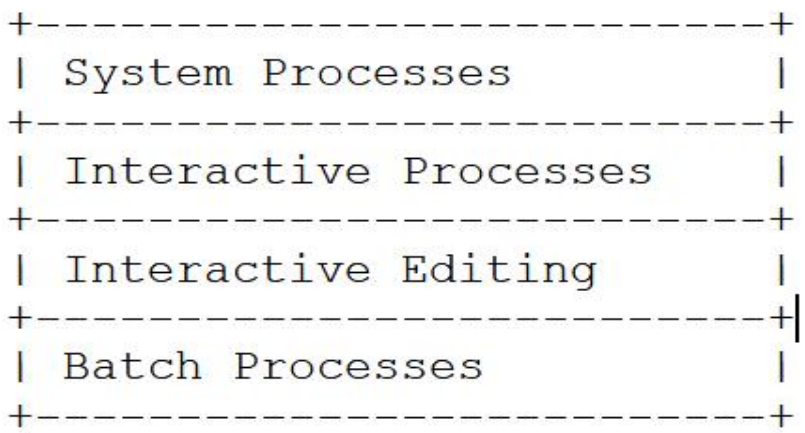
Disadvantages

- Too small a time quantum causes many context switches.
- Too large a time quantum makes it behave similarly to FCFS.

9. Multilevel Queue Scheduling

In **Multilevel Queue Scheduling**, the ready queue is divided into several separate queues based on process characteristics.

Example:



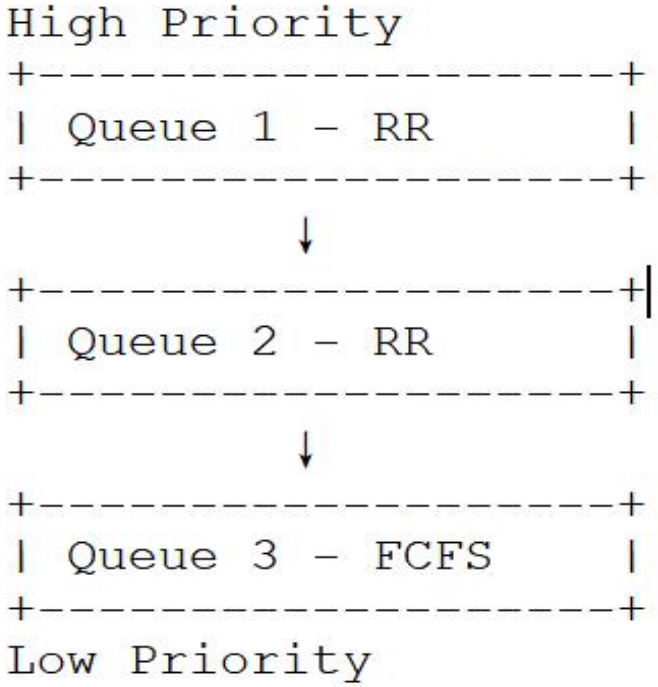
Each queue may use a different scheduling algorithm.

For example:

- Interactive queue → Round Robin
- Batch queue → FCFS

10. Multilevel Feedback Queue

A **Multilevel Feedback Queue (MLFQ)** allows processes to move between different queues depending on their behavior and CPU requirements.



A process that uses too much CPU time may be moved to a lower-priority queue, while processes that frequently wait for I/O may remain at a higher priority.

Advantages

- Flexible.
- Supports different types of processes.
- Can provide good response time.
- Helps balance interactive and CPU-intensive processes.

Disadvantages

- More complex to configure.
- Requires decisions about queue priorities and time quanta.

11. CPU Scheduling Criteria

Scheduling algorithms are evaluated using several criteria.

1. CPU Utilization

The percentage of time the CPU is busy.

Goal: Maximize CPU utilization.

2. Throughput

The number of processes completed per unit of time.

Goal: Maximize throughput.

CPU Scheduling Criteria

3. Turnaround Time

The total time from process submission to process completion.

Turnaround Time=Completion Time−Arrival Time

4. Waiting Time

The total time a process spends waiting in the ready queue.

Waiting Time=Turnaround Time−Burst Time

Waiting\ Time = Turnaround\ Time - Burst\ Time

CPU Scheduling Criteria

5. Response Time

The time between submitting a request and receiving the first response.

- **Response Time**=First CPU Start−Arrival Time

Response\ Time = First\ CPU\ Start - Arrival\ Time

6. Fairness

- Ensures that processes receive an appropriate share of CPU time and prevents indefinite waiting.

12. Important Scheduling Formulas

For examination purposes, remember:

Turnaround Time: **TAT**=CT−ATTAT = CT - AT

Waiting Time: WT=TAT−BTWT = TAT - BT

Response Time:

RT=First Start Time−ATRT = First\ Start\ Time - AT

Important Scheduling Formulas

Where:

CT = Completion Time

AT = Arrival Time

BT = Burst Time

WT = Waiting Time

TAT = Turnaround Time

RT = Response Time

13. Comparison of Scheduling Algorithms

Algorithm	Type	Main Principle	Main Problem
FCFS	Non-preemptive	First process to arrive runs first	Convoy effect
SJF	Non-preemptive	Shortest job runs first	Starvation
SRTF	Preemptive	Shortest remaining job runs first	More context switching
Priority	Both	Highest-priority process runs first	Starvation
Round Robin	Preemptive	Each process gets a time quantum	Quantum selection
Multilevel Queue	Usually both	Separate queues for process types	Inflexible
MLFQ	Preemptive	Processes can move between queues	Complex

Worked Example: FCFS

Consider the following processes:

Process	Arrival Time	Burst Time
P1	0	5
P2	1	3
P3	2	2

Gantt Chart

```
0         5         8         10
|---P1--|--P2--|--P3---|
```

Completion Time

- P1 = 5
- P2 = 8
- P3 = 10

Turnaround Time

P1=5−0=5P1 = 5-0 = 5 P2=8−1=7P2 = 8-1 = 7 P3=10−2=8P3 = 10-2 = 8

Waiting Time

P1=5−5=0P1 = 5-5 = 0 P2=7−3=4P2 = 7-3 = 4 P3=8−2=6P3 = 8-2 = 6

Average waiting time:

0+4+63=3.33\frac{0+4+6}{3} = 3.33

Average Waiting Time = 3.33 time units

Summary

- CPU scheduling is essential for efficient processor utilization and fair process execution. The most important algorithms to understand are FCFS, SJF, SRTF, Priority Scheduling, Round Robin, Multilevel Queue, and Multilevel Feedback Queue.
- For practical and examination questions, students should be able to construct Gantt charts and calculate completion time, turnaround time, waiting time, response time, and their averages.

Group assignment Questions (10 students and hand written)

1. Define CPU scheduling.
2. Explain the objectives of CPU scheduling.
3. Differentiate between preemptive and non-preemptive scheduling.
4. Explain FCFS with an example.
5. Explain SJF and its advantages and disadvantages.
6. Differentiate between SJF and SRTF.
7. Explain Priority Scheduling and the problem of starvation.
8. What is aging in CPU scheduling?
9. Explain Round Robin scheduling.
10. What is a time quantum?
11. Explain Multilevel Queue Scheduling.
12. Explain Multilevel Feedback Queue Scheduling.
13. Define waiting time, turnaround time, response time, and throughput.
14. Given a set of processes, construct a Gantt chart and calculate average waiting and turnaround times.
15. Compare FCFS, SJF, SRTF, Priority, and Round Robin scheduling algorithms.